

JsTrack

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Website : <https://github.com/lucademian/jsTrack>

License : GPL-3.0

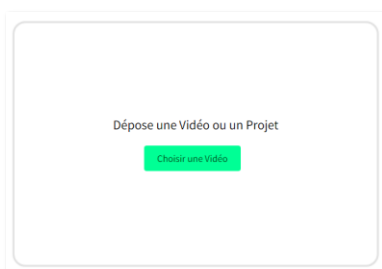
Adaptation : For mobile use and French translation in the PHYBOX

JsTrack is a javascript port of Tracker <https://physlets.org/tracker> which includes most of its features.

Designed to study Mechanics in Physic, JsTrack allows you to break down the trajectory of a moving object in order to characterize it.

Use :

Make a video of a moving object and import it into jsTrack.
In the "New Project" window, change the default settings if necessary.



Number of frames of video between each point of the trajectory

Number of points visible before and after the position where one is to avoid overloading

Nouveau Projet

Nom:

Nombre d'Images/Seconde:

d'images entre les Points:

Vitesse de la Vidéo:

Couleur des Axes:

Nombre de Points Placés Après:

Nombre de Points Placés Avant:

The two x and y axes, relative to which the distances of the points of the trajectory are measured. Position and orient it in a way that suits your project.

The scale allows you to match the distances on the screen with those in the experiment. Place its two ends next to something whose actual size you know, and double-click the value in the middle to change it.

The video playback bar to move from frame to frame or play the video

t(s)	x(m)	y(m)
0.099	0.40248	1.55422
0.165	0.40619	1.51409
0.198	0.4105	1.46701
0.231	0.41374	1.41918
0.264	0.41655	1.36292
0.297	0.41655	1.28977
0.33	0.4189	1.20724
0.363	0.42203	1.11817
0.396	0.42105	1.00634
0.429	0.42554	0.88959
0.462	0.42203	0.77179
0.495	0.41851	0.63992
0.528	0.42378	0.50277
0.561	0.42378	0.35508
0.594	0.42378	0.20738
0.627	0.42075	0.06728

The toolbar for using the various features of the application

The table of coordinates of each point you placed on the trajectory

THE TOOLBAR



Undo :

Undo or Redo an action, if you are not satisfied with the result.



Zoom :

Enlarge or decrease the video size to place points more precisely.



Move :

If video has been zoomed, it allows you to move it to the area where the object is located.



Video Size :

Returns to the original video size, without zooming.



Fullscreen :

Go full screen to work with more precision.



Motion :

Analyze the motion of an object by creating a new Trajectory. You can change the name and color (choose a color that stands out clearly against the background of your video).



Scale :

Add a scale to measure distances on your video. Click on two different places to make it appear and position it at a location where you know the actual distance. The distance between the two points is displayed : double-click on it to enter the value.



Add Points :

Using the playback bar, place your video at the moment the object begins to move. Click this button and tap where the object is to mark its position. The video then skips to the next frame to do the same with the following points.



Remove Points :

Touch a point to select it and delete it by clicking this button.



Export Image :

When you have finished positioning the points on the object's trajectory, click this button to send the image of the trajectory to your folder.



Export Coordinates :

Export to a spreadsheet file the table of coordinates of the points that appear in the panel on the right.



Save Project :

If you don't have time to finish this work in this session, you can save your project to continue it next time.



New Project :

If you have finished this work, start a new Project.

TRAJECTORIES



Displays or not the trajectory



Clear the trajectory

COORDINATES TABLE

This table shows the x,y coordinates of the points of the trajectory, measured in relation to your axes and taking into account the scale that you indicated.

When you click on a point to select it, it appears in green and its coordinates are highlighted in the table. Press the red cross if you want to delete this point.

t(s)	x(m)	y(m)	
0.099	-0.30768	1.67687	
0.198	-0.30768	1.60764	
0.231	-0.31153	1.5538	
0.264	-0.30384	1.48457	×
0.297	-0.30768	1.39227	
0.33	-0.30384	1.30381	
0.363	-0.29999	1.20766	
0.396	-0.29999	1.08458	
0.429	-0.29615	0.96151	

PLAYBACK BAR



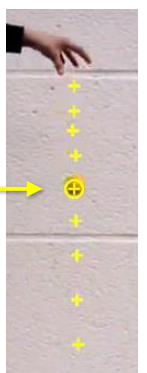
The small gray triangles on the left and right can be dragged to mark the beginning and end of the interesting part of your video during which the object is moving. The playback and analysis of your video will then only take place between these two points.

OTHER FEATURES

- Points Adjustment :

After you've placed your points along the object's path, you can readjust any that aren't positioned precisely enough. To do this, click the button again, then use the playback bar to go back frame by frame with ◀ and ▶. The point you chose to mark the position of the object at that moment is then surrounded by a yellow circle.

You can then click again in the center of the object to update the position of this point, the previous choice being replaced by the new one. But you can also move each point by hand to readjust it.



- Conversion to MP4 :

JsTrack can only play videos in MP4 format. If your video is in another format (for example, 3GP on Android), when you upload it, JsTrack will offer to convert it to MP4..